reasoning

artificial intelligence · prof. dr. nicolas meseth

1 thinking is writing
2 from a trick to a skill
3 learning from rewards
4 longer or more often
5 what it is not
6 why agents need it

some questions
are easier
on paper than
in your head.



image: ai-generated (gpt-image-2.5-flare)

part 1

thinking is writing

a tractor and a trailer cost 11,000 € together, and the tractor costs 10,000 € more than the trailer. what does the trailer cost?

500 €

1,000 €

1,500 €

10,000 €

live.hsos.uber.space/q

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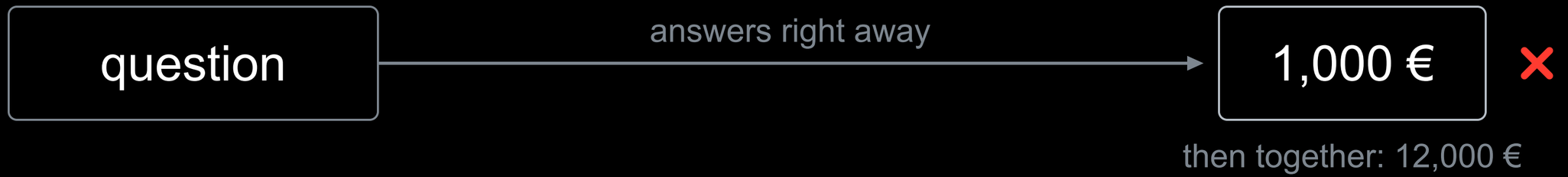
500 €

1,000 €

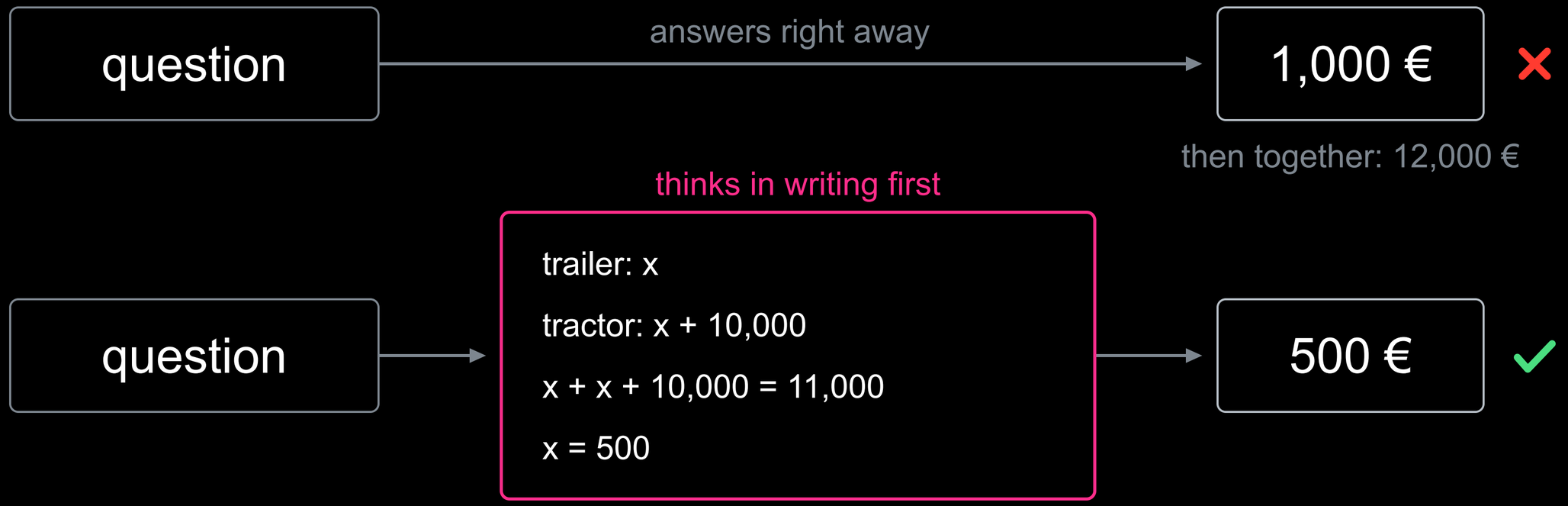
1,500 €

10,000 €

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then together: 12,000 €


question
answers right away
1,000 €
then together: 12,000 €
thinks in writing first
question
trailer: x
tractor: x + 10,000
x + x + 10,000 = 11,000
x = 500
500 €

a reasoning model is the same transformer predicting the same tokens, and what changed is what it has learned to write before the answer.

more tokens, more computation

every square is one token, and every token is one pass through all layers



500 tokens of thought before the answer are 500 passes

$$37 \times 48$$

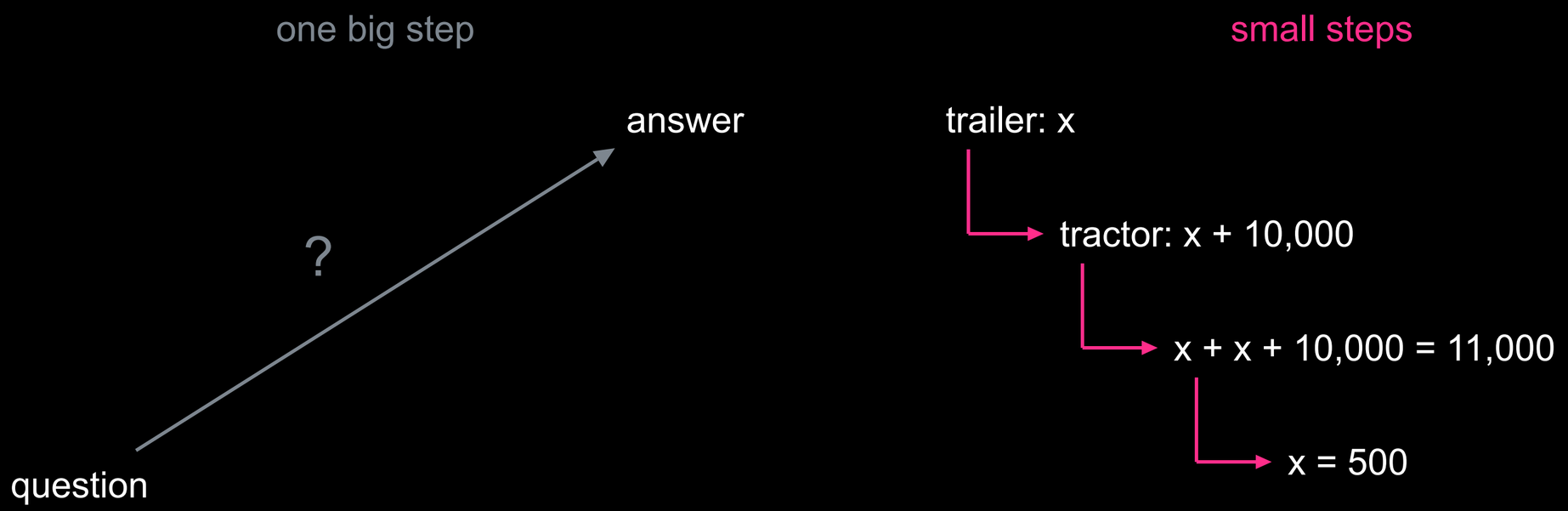
37 × 48

37 × 40 = 1,480

37 × 8 = 296

1,776

small steps are easier than one big step



```
trailer: 1,000
tractor: 11,000
together: 12,000
```

```
trailer: 1,000
tractor: 11,000
together: 12,000

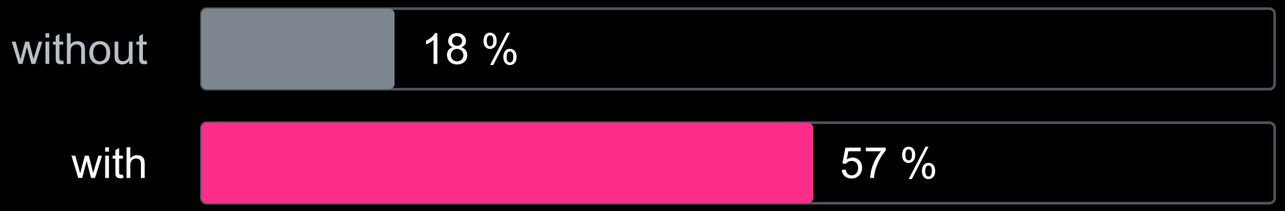
wait, that is 12,000, not 11,000.
trailer: x, tractor: x + 10,000
```

part 2

from a trick to a skill

asking for steps already helped

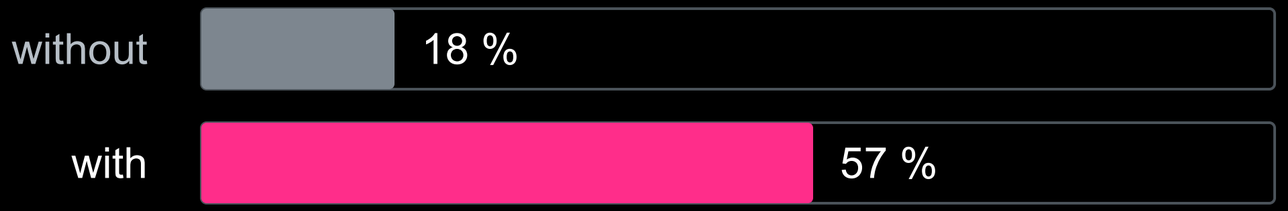
eight worked examples in the prompt: grade-school maths word problems



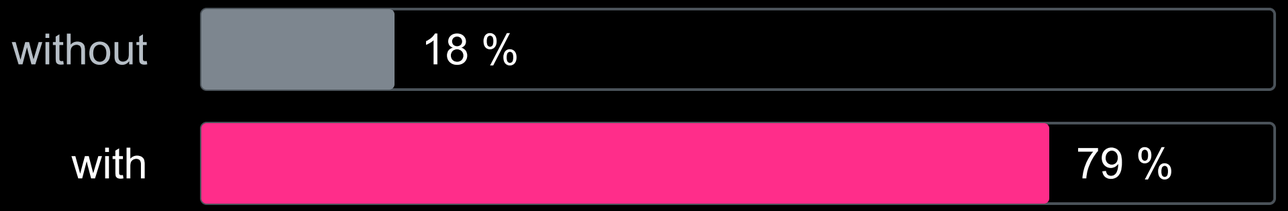
wei et al.: chain-of-thought prompting elicits reasoning, 2022 · kojima et al.: large language models are zero-shot reasoners, 2022

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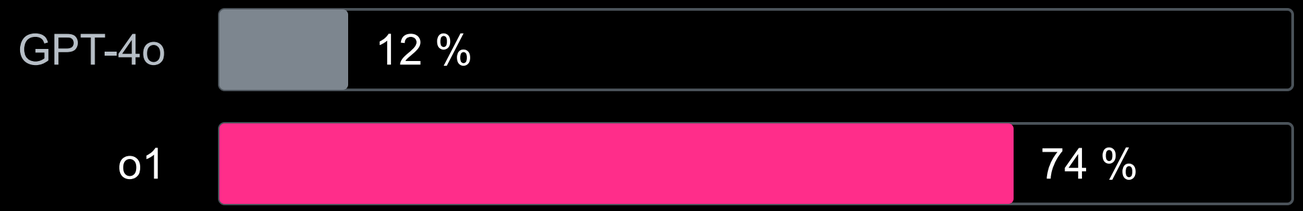
"let's think step by step" added to the question: arithmetic word problems



wei et al.: chain-of-thought prompting elicits reasoning, 2022 · kojima et al.: large language models are zero-shot reasoners, 2022

trained to think before every answer

aime, a maths competition for strong high-school students, one attempt per problem



openai: learning to reason with llms, 2024

asked for in the prompt

2021	scratchpads
01/2022	chain of thought
03/2022	self-consistency
05/2022	"let's think step by step"
10/2022	ReAct

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trained to reason	
09/2024	o1
12/2024	o3, Gemini 2.0 Flash Thinking
01/2025	DeepSeek-R1

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05/2022	"let's think step by step"	02/2025	Claude 3.7 Sonnet
10/2022	ReAct	07/2025	gold at the IMO
		08/2025	GPT-5, gpt-oss

part 3

learning from rewards

three stages, and only the last is new

1 pretraining

predict the next
token on books, code
and the internet

three stages, and only the last is new

1 pretraining

predict the next
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2 examples of thinking

optional:
a few thousand
worked answers

three stages, and only the last is new

1 pretraining

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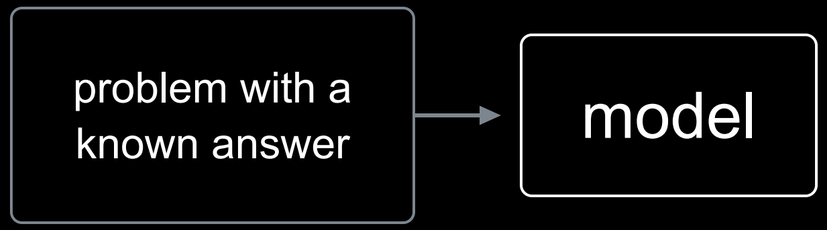
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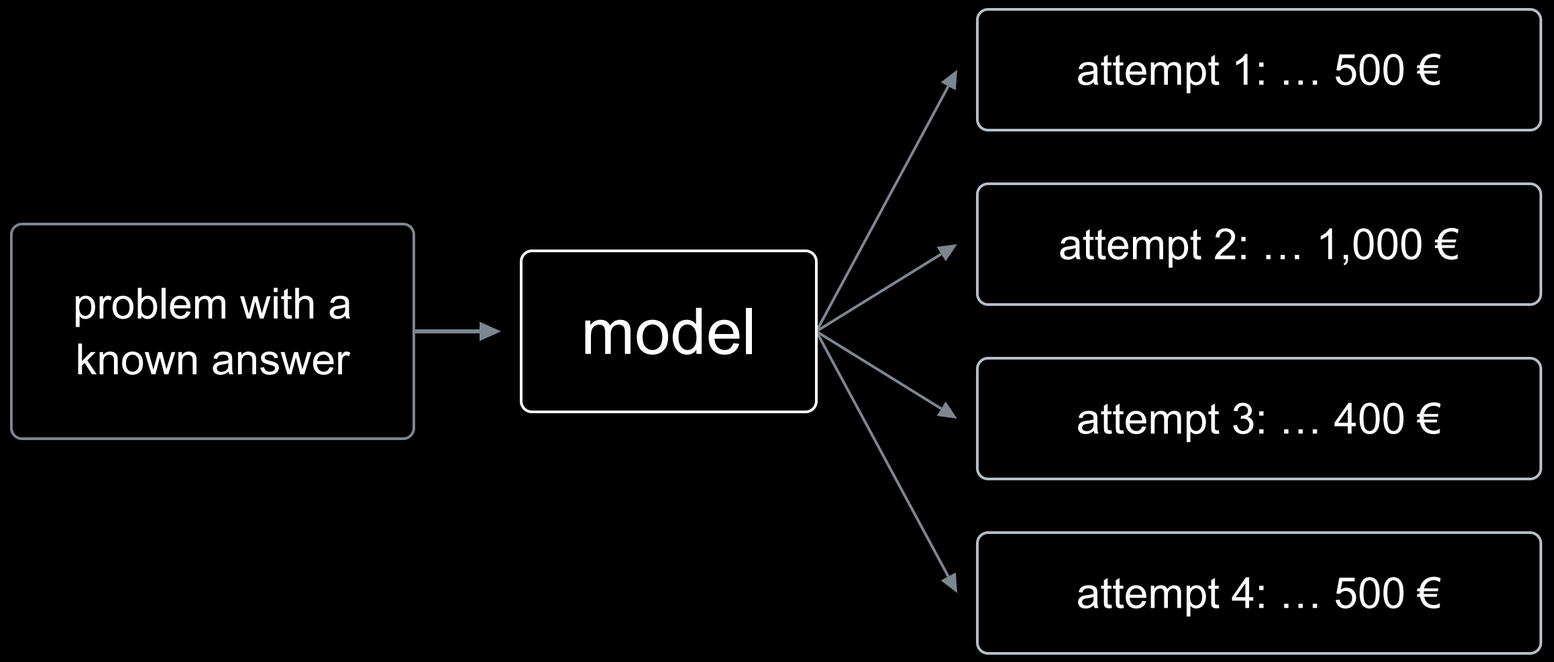
3 learning from rewards

attempts whose
answer a program
can check

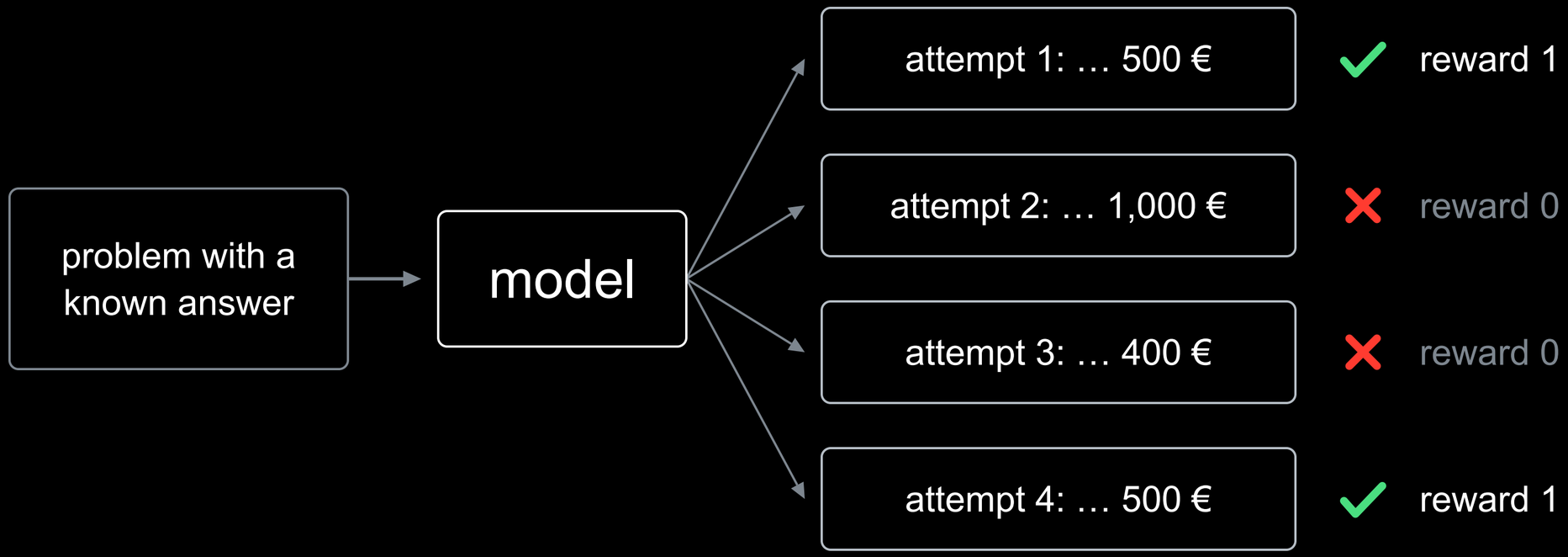
one round of learning from rewards



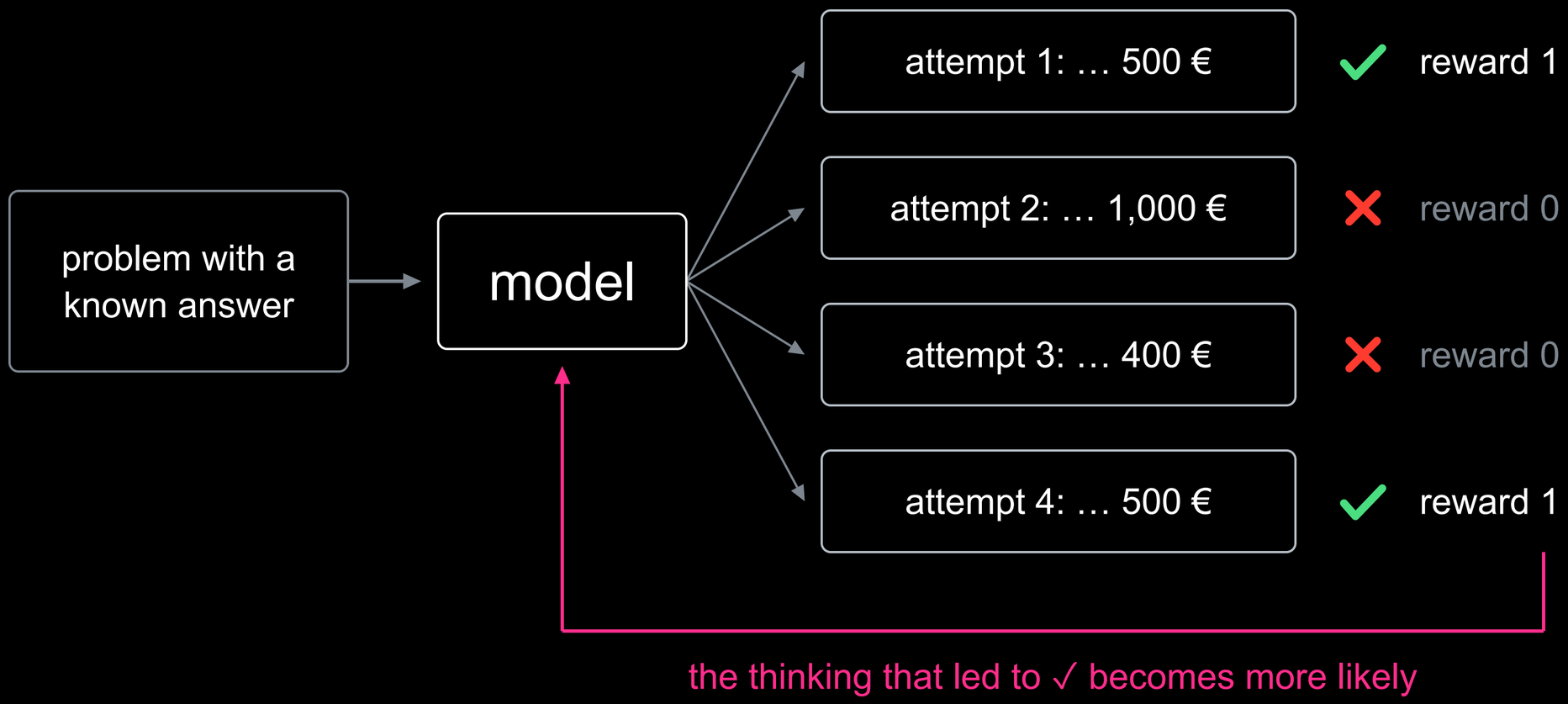
one round of learning from rewards



one round of learning from rewards



one round of learning from rewards



in learning from rewards, what does the program check?

every step of the thinking

only the final answer

how long the thinking is

whether it sounds human



live.hsos.uber.space/q

in learning from rewards, what does the program check?

- every step of the thinking
- only the final answer
- how long the thinking is
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live.hsos.uber.space/q

in learning from rewards, what does the program check?

every step of the thinking

only the final answer

how long the thinking is

whether it sounds human



live.hsos.uber.space/q

wait, wait. wait. that's an aha moment i can flag here.

DeepSeek R1-Zero, trained with rewards alone, improved on AIME from 16 % to 71 %.

deepseek-ai: deepseek-r1: incentivizing reasoning capability in llms via reinforcement learning, 2025

a program can check

- ✓ maths: compare with the result
- ✓ code: run the tests

why maths and code came first

a program can check

- ✓ maths: compare with the result
- ✓ code: run the tests

a loophole

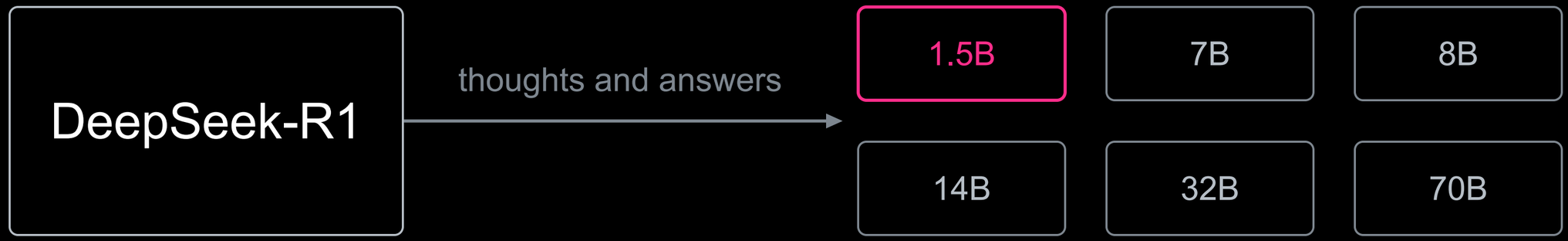
the test only checks that it runs

↓

a program that runs and does nothing

reward hacking

from a large model to your laptop



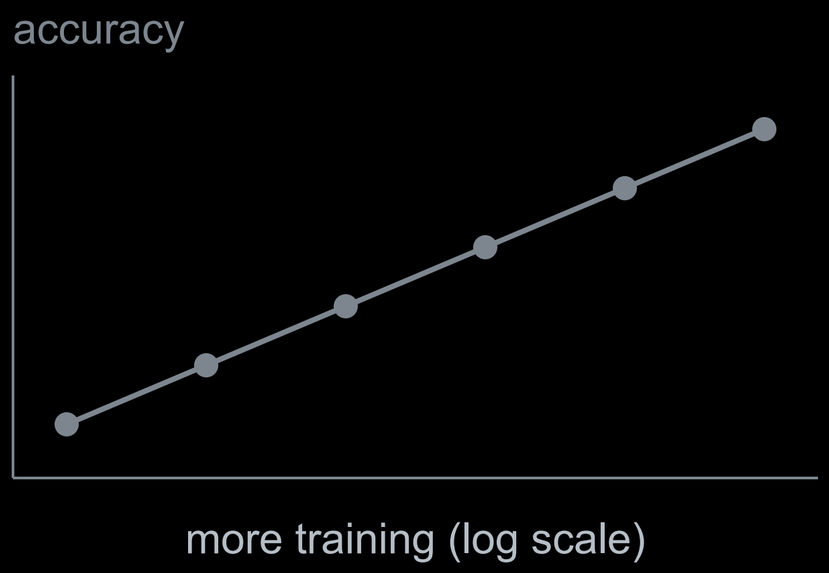
even the smallest one beat GPT-4o on AIME

deepseek-ai: deepseek-r1, 2025

part 4

longer or more often

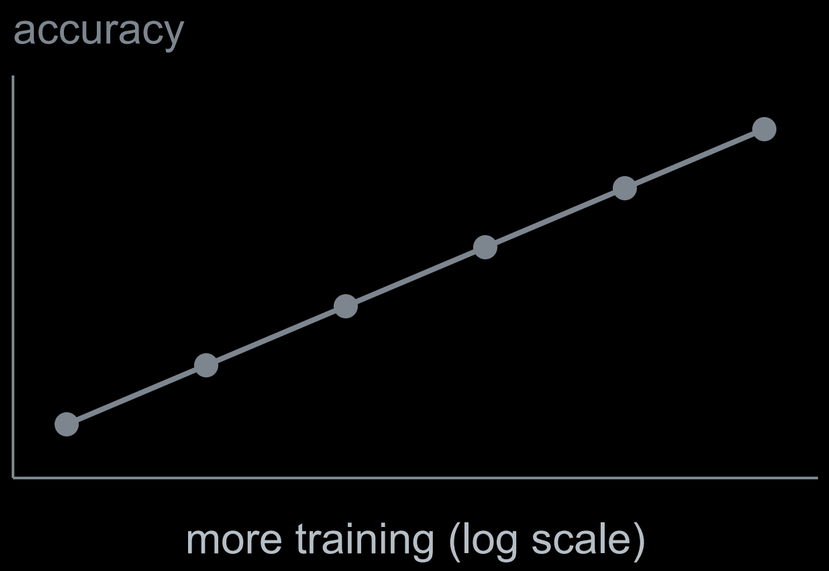
two curves

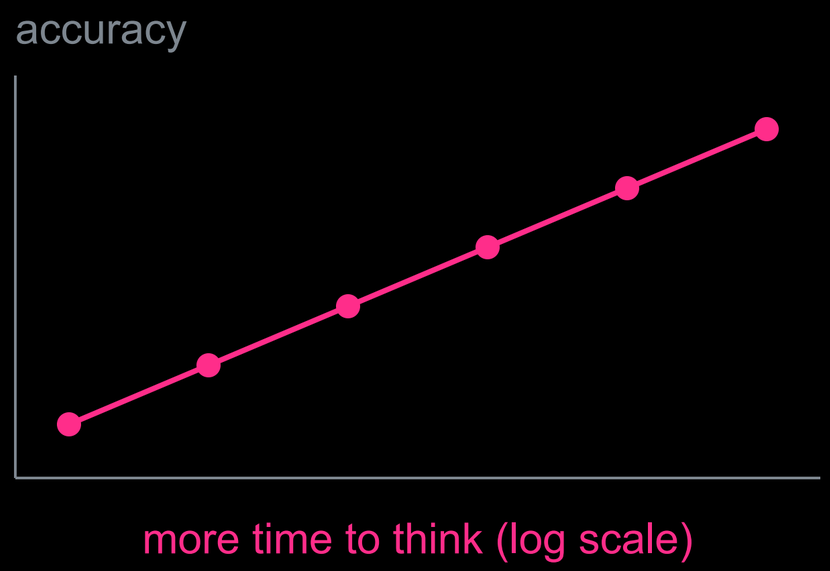


shape after openai: learning to reason with llms, 2024; curves for illustration

longer or more often

two curves





shape after openai: learning to reason with llms, 2024; curves for illustration

longer or more often

think longer, or think more often

think longer

step 1

step 2

wait, step 2 is wrong

step 2, again

step 3

answer: 500 €

longer or more often

think longer

step 1

step 2

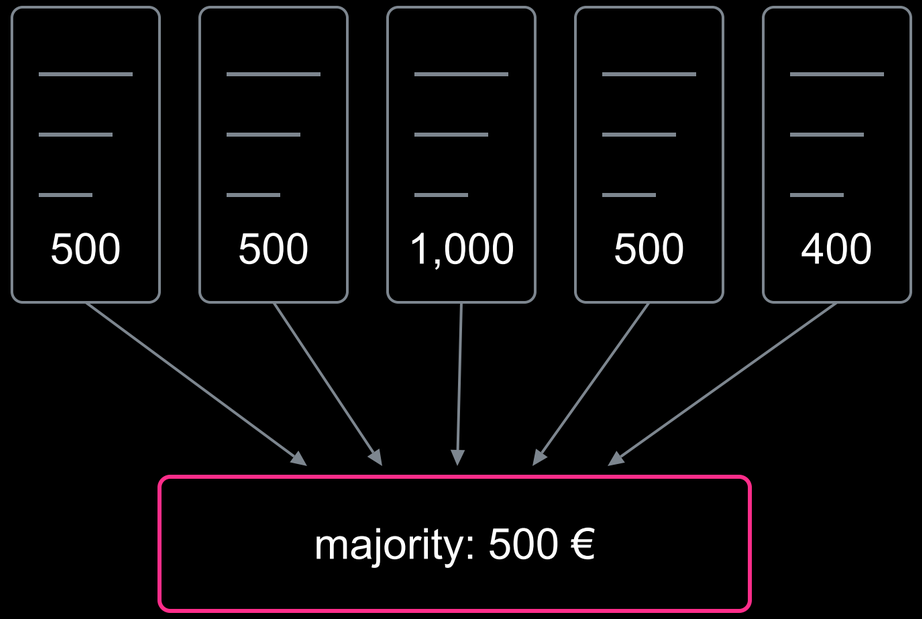
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step 2, again

step 3

answer: 500 €

think more often



think longer, or think more often

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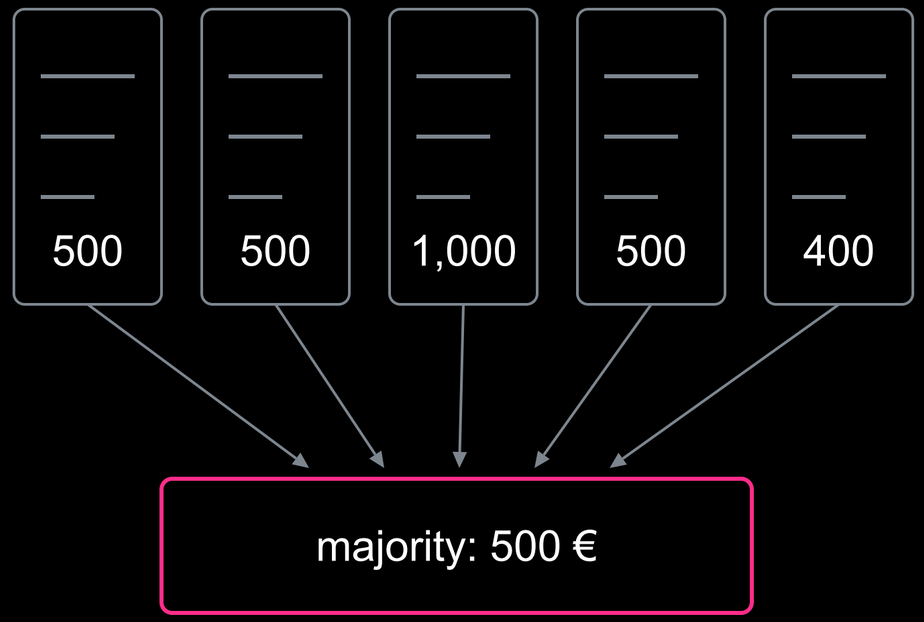
wait, step 2 is wrong

step 2, again

step 3

answer: 500 €

think more often



o1 on AIME: 74 % with one attempt, 83 % with 64 attempts and a majority vote

learning and search are the two methods that keep scaling with computation, and a reasoning model uses both.

thinking is not free

76 % about $26 per puzzle

88 % 172 × the computation, thousands of dollars per puzzle

o3 on ARC-AGI, puzzles that are easy for people; thinking tokens are billed like answer tokens.

arc prize: openai o3 breakthrough high score on arc-agi-pub, 2024

longer or more often

low or high

change the numbers of the tractor question.

in LM Studio with gpt-oss-20b, the setting is under Reasoning in the panel on the right.

longer or more often

low or high

change the numbers of the tractor question.

ask once with reasoning set to low and once with high.

in LM Studio with gpt-oss-20b, the setting is under Reasoning in the panel on the right.

longer or more often

low or high

change the numbers of the tractor question.

ask once with reasoning set to low and once with high.

compare the time, the number of tokens and the thoughts.

in LM Studio with gpt-oss-20b, the setting is under Reasoning in the panel on the right.

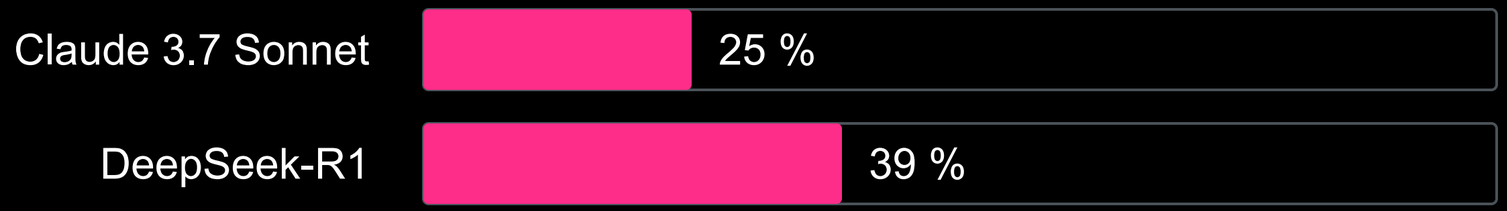
longer or more often

part 5

what it is not

the thoughts are not always the reason

used a hint to the answer and said so in its thoughts



anthropic: reasoning models don't always say what they think, 2025

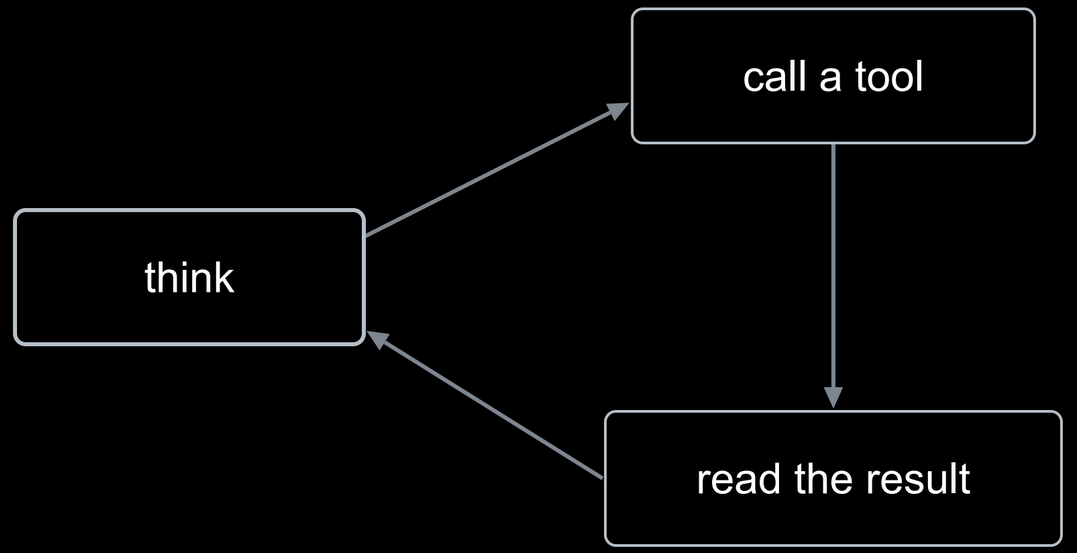
a long and convincing chain of thought is no proof, so the answer still has to be checked.

shojaee et al.: the illusion of thinking, 2025, and the reply the illusion of the illusion of thinking

part 6

why agents need it

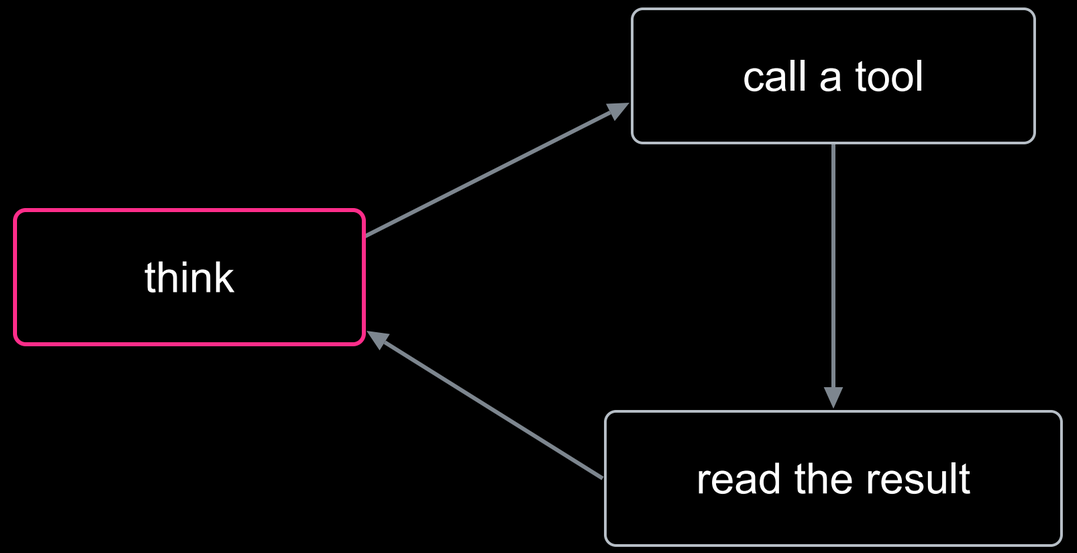
thinking inside the loop



a plan before the first step

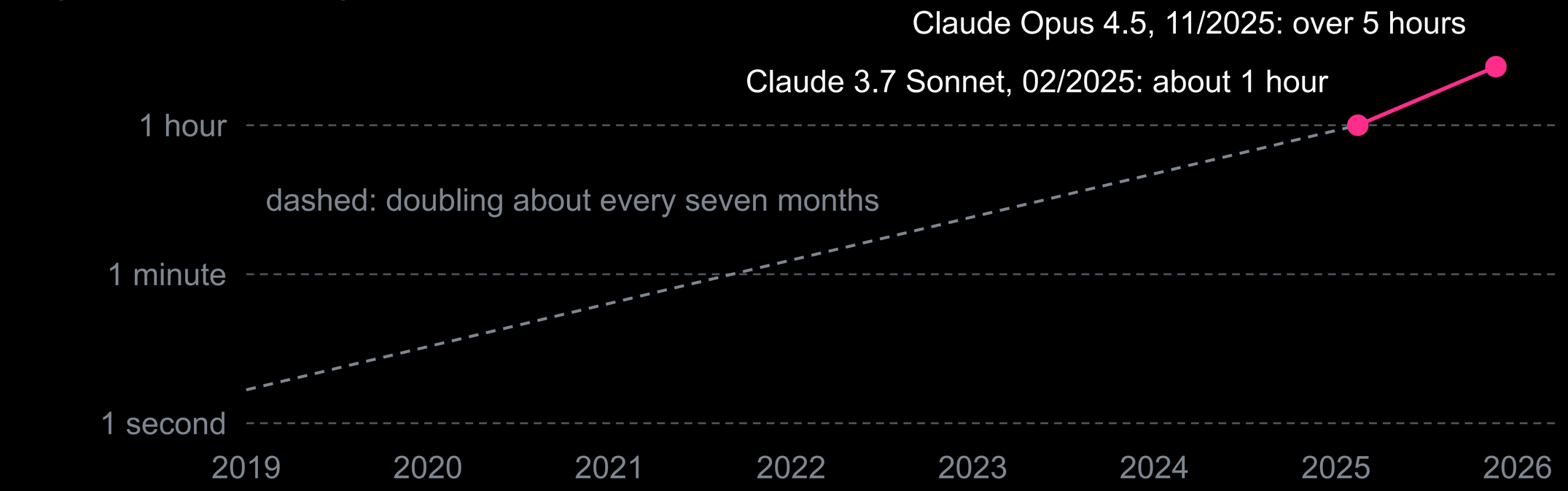
a check after each result

wait, that failed



how long an agent can work

length of the tasks an agent solves in half of the cases



metr: measuring ai ability to complete long tasks, 2025, and time horizon 1.1, 2026

an agent that works for hours is a model that thinks before it acts, many thousands of times in a row.

j. wei et al.: chain-of-thought prompting elicits reasoning in large language models, 2022

t. kojima et al.: large language models are zero-shot reasoners, 2022

openai: learning to reason with llms, 2024

deepseek-ai: deepseek-r1: incentivizing reasoning capability in llms via reinforcement learning, 2025

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